Crater and Boulder Detection on Lunar Surfaces using YOLOv8s

Title

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1. Introduction

Space exploration presents countless challenges - one of them is the automated analysis of celestial surfaces. On the Moon, detecting craters and boulders plays a critical role in mission planning, navigation, and scientific research. Traditional manual annotation of such features is time-consuming and error-prone.

This project, part of the Lunar Vision Challenge, aims to address this problem using a deep learning-based object detection pipeline. The focus was to train and deploy a lightweight Convolutional Neural Network (CNN), YOLOv8s, to detect craters and boulders on grayscale lunar imagery with high accuracy and efficiency.

The goal was not only to build an effective model but also to make the process simple to reproduce and efficient to run, even with limited computational power.

2. Problem Statement

Given a dataset of lunar surface images, the task is to detect and localize two types of objects:

- Crater (Class 0)
- Boulder (Class 1)

Each image should be processed to generate YOLO-style label files, where each line in a .txt file represents an object: [class_id, x_center, y_center, width, height].

Performance is judged on how accurately the predicted labels match the ground truth in terms of precision, recall, and mean average precision (mAP).

3. Methodology

Model Architecture:

The model used is YOLOv8s, the small variant of the YOLOv8 object detection framework developed by Ultralytics. This version is known for its balance of speed and accuracy, making it ideal for use in constrained environments like Kaggle notebooks.

Dataset:

Only the official Lunar Vision Challenge dataset was used. No external data or synthetic samples were added.

Training Details:

- Model: YOLOv8s
- Epochs: 50
- Augmentation: Default
- Platform: Kaggle GPU
- Batch Size: 16
- Loss: Standard YOLO loss
- Optimizer: SGD

Inference:

After training, inference was performed on test images using the final checkpoint. The output was a set of YOLO label .txt files.

4. Results

Evaluation metrics from the validation set were as follows:

- Precision: 0.84

- Recall: 0.81
- mAP@0.5: 0.897
- mAP@0.5:0.95: 0.735
- Fitness Score: 0.75

The model struck a strong balance between recall and precision. Most of the missed detections were small craters near larger boulders.

5. Innovation & Visualizations

Though we did not implement architectural innovations due to resource constraints, we used Grad-CAM to visualize where the model focuses when detecting craters.

Grad-CAM helps verify that the model focuses on meaningful features, providing insight into the CNN's decision-making process. A Gradio UI was developed to easily visualize these results.

6. Key Challenges

Computational Constraints:

Training a deep learning model is compute-intensive. We used Kaggle Notebooks with T4 GPUs which imposed limits.

To work around this:

- Chose YOLOv8s
- Trained for 50 epochs
- Used default augmentations

Debugging and Visualization:

Due to a lack of raw test images, we relied on Grad-CAM and internal validation to understand the model's

behavior.

7. Folder Structure

LunarVisionSubmissions/

model/ (best.pt)

data.yaml

predictions/ (*.txt)

code/

train.py

infer.py

gradio_ui.py

innovation/

gradcam_visualize.py

gradcam_samples/*.jpg

README.md

final_submission.pdf

8. Conclusion

Despite limitations in compute and time, this project successfully demonstrated the use of YOLOv8s to detect craters and boulders on lunar imagery. The pipeline is lightweight, interpretable, and deployable. Future improvements could include using YOLOv8m, better augmentation, and temporal modeling.

9. References

1. Ultralytics YOLOv8 - https://github.com/ultralytics/ultralytics

2. YOLOv8 Paper - https://arxiv.org/abs/2304.00501

3. Grad-CAM - https://arxiv.org/abs/1610.02391

4. Lunar Vision Dataset

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